Week 2

1-G-Coin Problem

```
#include<stdio.h>
int main(){
    int n;
    int coins[]={1000,500,100,50,20,10,5,2,1};
    int count=0;
    scanf("%d",&n);
    for(int i=0;i<9;i++){
        while(n>=coins[i]){
            n=n-coins[i];
            count++;
        }
    }
    printf("%d",count);
    return 0;
}
```

2-G-Cookies Problem

```
#include<stdio.h>
int main(){
    int n,m,i,j,ans=0;
    scanf("%d",&n);

    int g[n];
    for(i=0;i<n;i++)
        scanf("%d",&g[i]);

    scanf("%d",&m);

    int s[m];
    for(i=0;i<m;i++)
        scanf("%d",&s[i]);

    for(i=0;i<n;i++)
        for(j=0;j<m;j++)
            if(s[j]>=g[i]){
                ans++;
                s[j]=0;
                break;
            }

    printf("%d",ans);
    return 0;
}
```

3-G-Burger Problem

```
#include <stdio.h>
#include <stdlib.h>

int compare(const void *a, const void *b) {
    return (*(int *)b - *(int *)a);  // Descending order
}

int main() {
    int n;
    scanf("%d", &n);

    int calories[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &calories[i]);
    }

    // Greedy: eat burgers in decreasing order of calories
    qsort(calories, n, sizeof(int), compare);

    long long total = 0;
    long long power = 1;  // 3^0

    for (int i = 0; i < n; i++) {
```

```
        total += calories[i] * power;
        power *= n;
    }

    printf("%lld\n", total);

    return 0;
}
```

4-G-Array Sum max

```
#include <stdio.h>
#include <stdlib.h>

int cmp(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}

int main() {
    int n, i;
    long long sum = 0;

    scanf("%d", &n);

    int a[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    qsort(a, n, sizeof(int), cmp);

    for (i = 0; i < n; i++)
        sum += (long long)a[i] * i;

    printf("%lld", sum);

    return 0;
}
```

5-G-Product of Array elements-Minimum

```
#include <stdio.h>
#include <stdlib.h>

int asc(const void *a, const void *b) {
    return *(int *)a - *(int *)b;
}

int desc(const void *a, const void *b) {
    return *(int *)b - *(int *)a;
}

int main() {
    int n, i;
    long long sum = 0;

    scanf("%d", &n);

    int a[n], b[n];

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++)
        scanf("%d", &b[i]);

    qsort(a, n, sizeof(int), asc);
    qsort(b, n, sizeof(int), desc);

    for (i = 0; i < n; i++)
        sum += (long long)a[i] * b[i];

    printf("%lld", sum);

    return 0;
}
```